

Aerodynamic consultant and Design – test engineer

ASU Formula SAE- Sun Devil
Motorsports(SDM)

Aerodynamics team

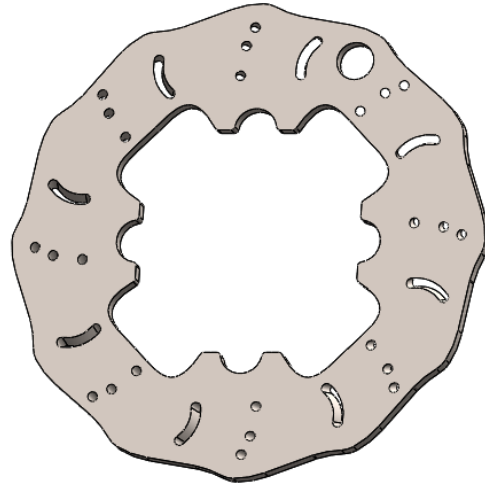
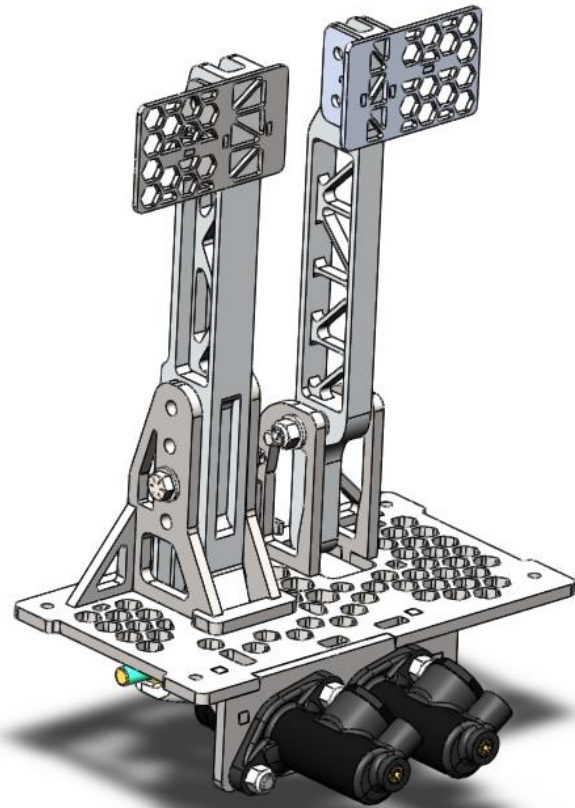
Chassis team

Manufacturing team

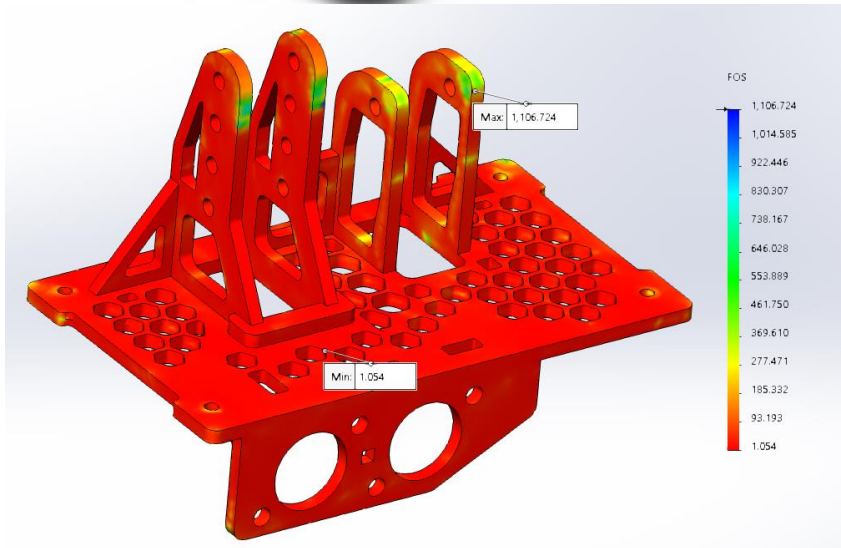
Powertrain team

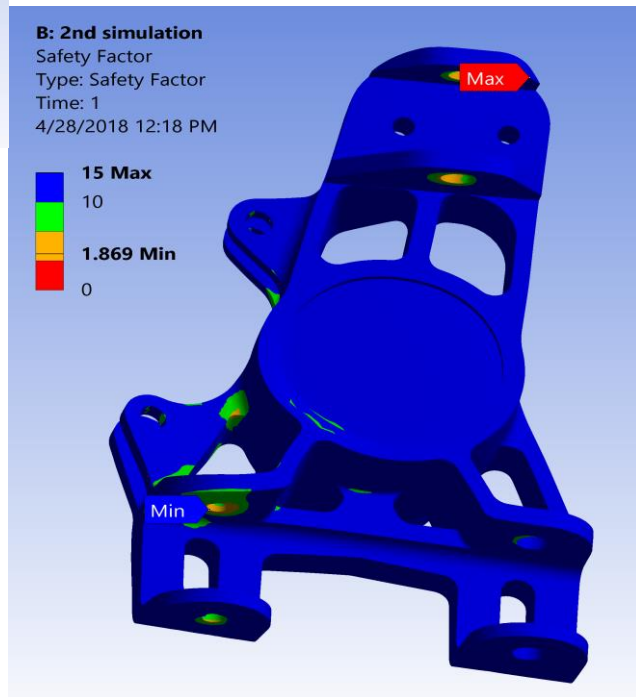
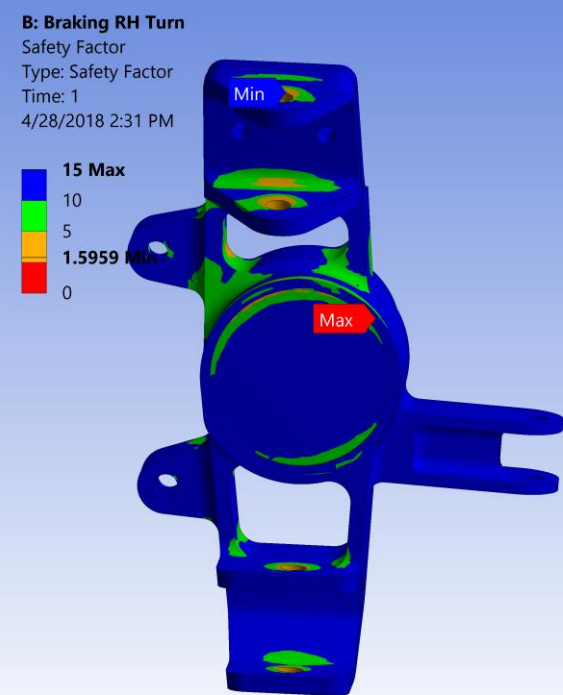
August 2017- May 2019





- Rotor has holes, slots, and scalloped edge for increased heat transfer and weight reduction
- Angled features are added to act like a fan blade to promote heat transfer
- Pedal tray was designed to be made out of one sheet of $\frac{1}{4}$ " steel, assembled with tongue and groove, and welded.
- Figure: Top left- Final Pedal Box Assembly.
- Figure: Top right: Final Brake Rotor Design
- Figure: Bottom: Optimized Design final FOS came out to be 1.





- 6061 – T651 material used.
- Refined upright design to save weight and achieve higher FOS.
- Improvements made to each component either in increased FOS, decreased weight, or simplified manufacturability.
- Suspension is designed concurrently with uprights and changes often during the design process.
- Figure Top: Initial upright design.
- Figure bottom: Final Design



- In charge of cost analysis and material selection with cross functional teams including operations and EHS teams. Materials selected considering finances and supplier (composite envisions and West Systems).
- Analysis considering the financially viable fabric options to select the best option.
- Utilized West Systems 105 resin and 207 special clear hardener with metering pumps for adhesive mixing.
- Figure: Top- Finished SDM 19 car.
- Figure: Bottom right: Twill 3k fiber-fabric, Bottom left: Biaxial fiber-fabric
- For the SDM 18 car being manufactured, a single layer of 2*2 twill biaxial 3k followed by a layer of 2*2 twill 3k.
- For 2019, 2*2 twill 3k was the layer utilized.

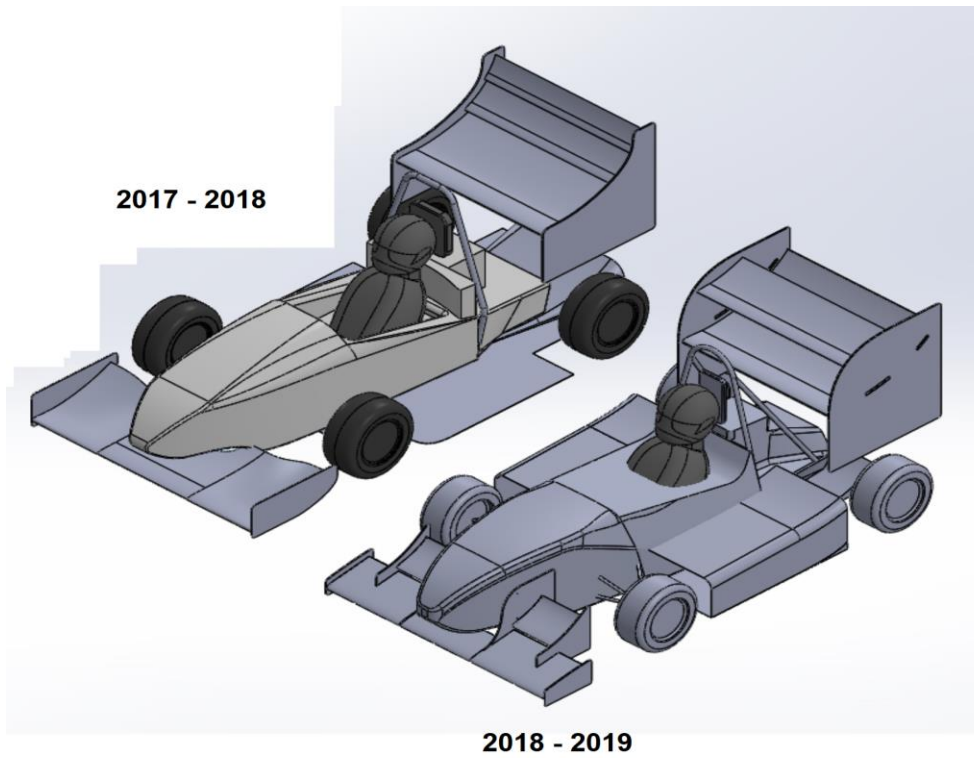




- In charge of performing wet composite layups for SDM 19 using 2 part epoxy (resin + hardener).
- In charge of Design for Manufacturing for SDM 19 to account for failures in SDM 18 manufacturing.
- Also utilized one part cyanoacrylates and methacrylates for adhesion functions (bonding mounting tabs to rear wings end ribs). And also utilized PVA (PolyVinyl Acetate) films for bonding to glass.
- Figure: Left-The vacuum bagged sidepod after layup. Right- The vacuum bagged vehicle undertray with aluminium hexcore underneath sidepod supports.



- Utilized a fiberglass split mold to perform the nosecone layup (female mold).
- CSM Fiberglass mold created using a wooden nosecone mold (male mold). Split mold created using half of the wooden mold and then fastened together.
- All layup performed in accordance with safety regulation while handling epoxy.
- Figure: Left- The demolded nosecone before cutting.
- Figure: Right – Bunny suit used during wet layups for the inside half of the nosecone.



- Collaborated with the chassis team during the assembly of the vehicle.
- Performed pre-treatment before adhesion. Eg: Scuffing up the surface for bonding purposes. Similar treatment utilized before performing welding operations to expose base metal.
- Performed welding operations and surface treatment for the tubular space frame assembly with chassis.
- Used Arc welding and MIG welding for welding purposes.
- Figure: Top- Aero package update from SDM 18 vehicle to the SDM 19.
- Figure: Bottom- The composite nose cone after cutting ready for assembly.

Production and Operations Support for SDM and SDR Manufacturing team



Performed hands on wet layup for the Nosecone, Undertray and wings in accordance with safety sheet and regulations and documented them for future prototypes.



Implement Lean principles (5S, Visual Management, Facility Layout). Analyzed the possible failures in SDM 19 design and production from failure data from SDM 18,17,16



Providing hands on production, tooling and machining support to the manufacturing team in SDM 19 and 18 assembly process.



SDM 19: Designed molds, mounts and fixtures for the wings, undertray and nosecone and completed wet layups for FR composites, CNC machining for mold assembly.



SDM 18: Apart from Body layups, Completely assembled seat mold from scratch for layup and completed wet layups during manufacturing of the SDM 18 car in accordance with OSHA regulations



Composite Consultant with the Sun Devil Racing(BAJA SAE) team where I consulted with the team for transition from Al sheet body panels to Composites for innovation. For initial step, implemented Kevlar/Carbon hybrids for the belly pan.